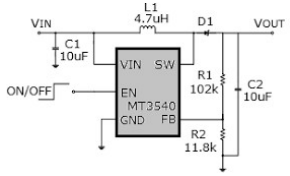
noknok 1.42 inch TFT Display
80x160 pixel
IPS panel



Voltage booster for Panel Backlight to 4.5V

+3.3V-IN
L1
Inductor 4.7uH
U2
MT3540-F23
IN SW
EN GND FB
D3
D_Schottky
+4.5V
VOUT
C7 10uF
C5 0.1uF
R3 33k
R4 12k
C6 22uF
GND

PWR_FLAG
+3.3V-IN
Avoid voltage overshoot when inserting power
R1 10R
D2
D_Zener
GND
+3.3V-PROT
Bulk Decoupling Capacitors
C1 10uF
C2 0.1uF
GND
no Pull-up Resistors for I2C, as they are entirely on the host
if module based pull-ups are required add e.g. 10k Ohm from 3.3V-prot to SDA and 10k Ohm to SCL.
Decoupling Capacitor for MCU only needed if not too close to Bulk decoupling caps

Qwiic / Stemma QT Connectors
J2
JST 4 Pin Qwiic I2C Connector
J1
JST 4 Pin Qwiic I2C Connector
GND

U1
CH32V003F4U6
PC0 PC1 PC2 PC3 PC4 PC5 PC6 PC7
PD0 PD1 PD2 PD3 PD4 PD5 PD6 PD7
PA1 PA2
VDD
VSS

Status Indicator LED
+3.3V-PROT
D4
LED
R5 2.2k
GND

+3.3V-PROT
Firmware Flashing
Footprint incl. M2.5 mounting hole
VCC
SWIO
GND
J3
noknok_FlashPads_I2C-module
GND

+4.5V
2N7002 or AO3400A (pin2pin)
Q1
33R R8
R9 100k
R10 100R
GND
Backlight and brightness control

GND LEDA LEDK VCC NC GND
CS RES RS/DC SCL/SCK SDA/MOSI NC_TOUCH2 NC_TOUCH1
22R R6
22R R7
noknok_ER-TFT1.42-1
DS1
GND

GND
GND TX RX
J4
noknok_UART_debug_1x3_pogo

Reset Crictuit
Delays start of MCU.
+3.3V-PROT
R2 10k
C4 0.1uF
GND

Pin suggestions:
D0_SCLK to PC5 (MUST)
C1_MOSI to PC6 (MUST)
RES# to any free GPIO
D/C# to any free GPIO

+3.3V-PROT
C8 0.1uF
C9 1uF
GND
Decoupling for Panel

H1 noknok_MountingHole
H2 noknok_MountingHole
H3 noknok_MountingHole

https://noknok.app
Sheet: /
File: module-I2C-Display_1_42inch.kicad_sch
Title: 1.42 incl Display on CHV32003F4U6
Size: A4 | Date: 2026-08-25 | Rev: 2.0
KiCad E.D.A. 10.0.0 | Id: 1/1